

Bachelor's Thesis Assignment

Institute: Institute of Solid Mechanics, Mechatronics and Biomechanics
Student: **Jakub Hloušek**
Degree program: Mechatronics
Branch: no specialisation
Supervisor: **Ing. Michal Bastl, Ph.D.**
Academic year: 2025/26

As provided for by the Act No. 111/98 Coll. on higher education institutions and the BUT Study and Examination Regulations, the director of the Institute hereby assigns the following topic of Bachelor's Thesis:

Embedded Implementation of an Attendance System for Integration into an Enterprise Application

Brief Description:

The student will design and implement an attendance system based on an embedded platform, integrated with the company's internal software for work time recording. The system will use identification methods (such as RFID, NFC, or PIN code) to record employee arrivals and departures. The recorded data will be processed by the designed unit and transmitted to the company software through an appropriate communication interface. The thesis will include the design of the hardware and software architecture of the system, implementation of a functional prototype, and verification of its functionality in a model office environment.

Bachelor's Thesis goals:

1. Conduct a review of available hardware and software solutions for attendance systems and analyze possibilities for their integration with enterprise systems.
2. Design the architecture of the attendance system according to specific requirements for integration into the company's software.
3. Implement a prototype including an identification module (RFID/NFC/PIN) and a communication interface with the company's software.
4. Verification of the functionality of the solution in a model office environment and evaluation of its potential for practical deployment.

Recommended bibliography:

- [1] VALÁŠEK, Michael. Mechatronika. Praha: České vysoké učení technické, 1995. ISBN 80-01-01276-X.
- [2] HEROUT, Pavel. Učebnice jazyka C. 1. vydání. Brno: Computer Press, 2004. ISBN 80-251-0452-3

[3] MALÝ, Martin. ESP32 prakticky : od základních obvodů k pokročilým aplikacím. 1. vydání. Praha : CZ.NIC, z.s.p.o., 2024. 525 s. ISBN 978-80-88168-79-9.

Deadline for submission Bachelor's Thesis is given by the Schedule of the Academic year 2025/26

In Brno,

L. S.

prof. Ing. Jindřich Petruška, CSc.
Director of the Institute

doc. Ing. Jiří Hlinka, Ph.D.
FME dean